

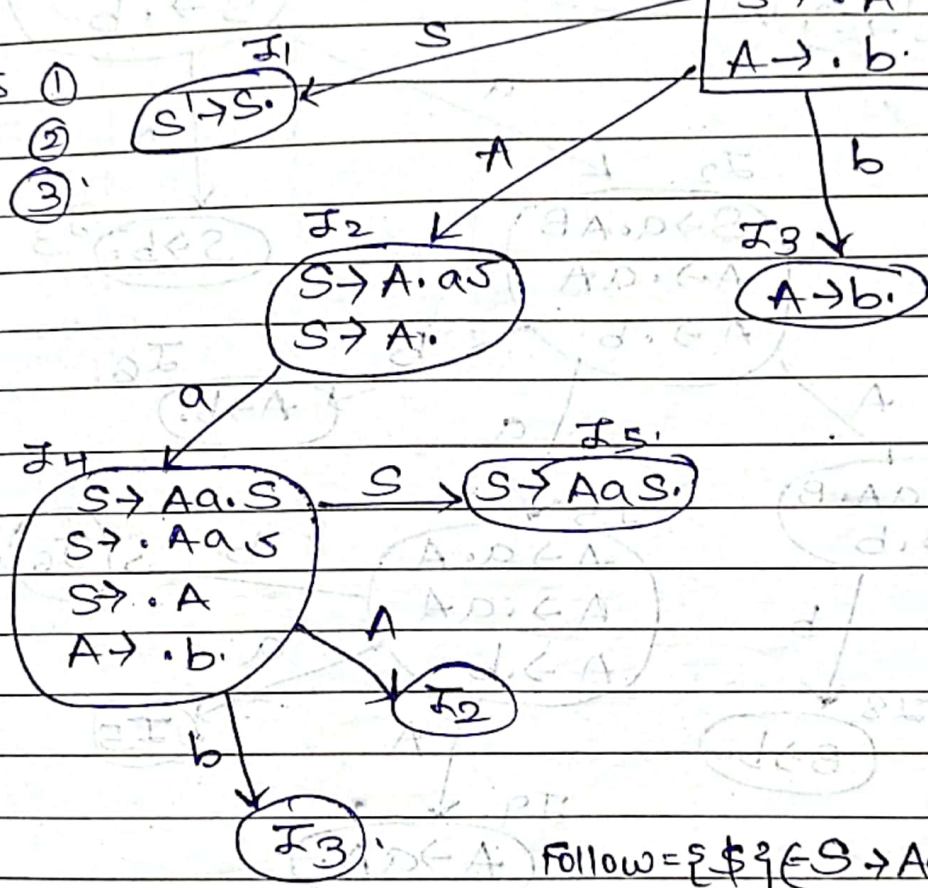
* SLR(1)

$S \rightarrow AaS / A$
 $A \rightarrow b$

Augmented Grammar.

$\mathcal{I}_0$ $\begin{cases} S' \rightarrow \cdot S \\ S \rightarrow \cdot AaS \\ S \rightarrow \cdot A \\ A \rightarrow \cdot b \end{cases}$

- $S \rightarrow AaS$ ①
- $S \rightarrow A$ ②
- $A \rightarrow b$ ③



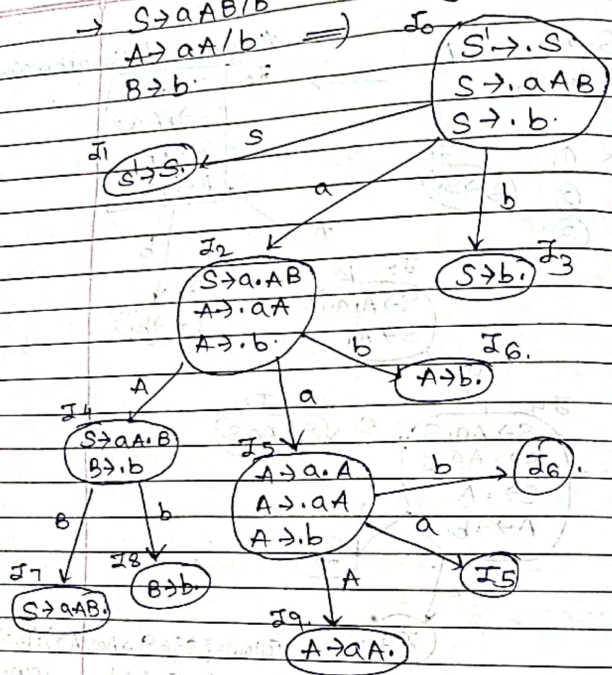
Follow = { \$ } ($S \rightarrow AaS / A \Rightarrow F(S) = \{b\}$)

Follow = { a, \$ } ($A \rightarrow b \Rightarrow F(A) = \{b\}$)

	a	b	\$	S	A
$\mathcal{I}_0$		$\mathcal{I}_3$		1	2
$\mathcal{I}_1$			Accept.		
$\mathcal{I}_2$	$\mathcal{I}_4$				
$\mathcal{I}_3$	$\mathcal{I}_3$		$\mathcal{I}_3$		
$\mathcal{I}_4$		$\mathcal{I}_3$		5	2
$\mathcal{I}_5$			$\mathcal{I}_1$		

Augmented Grammar

$S \rightarrow aAB/b.$
 $A \rightarrow aA/b.$
 $B \rightarrow b.$



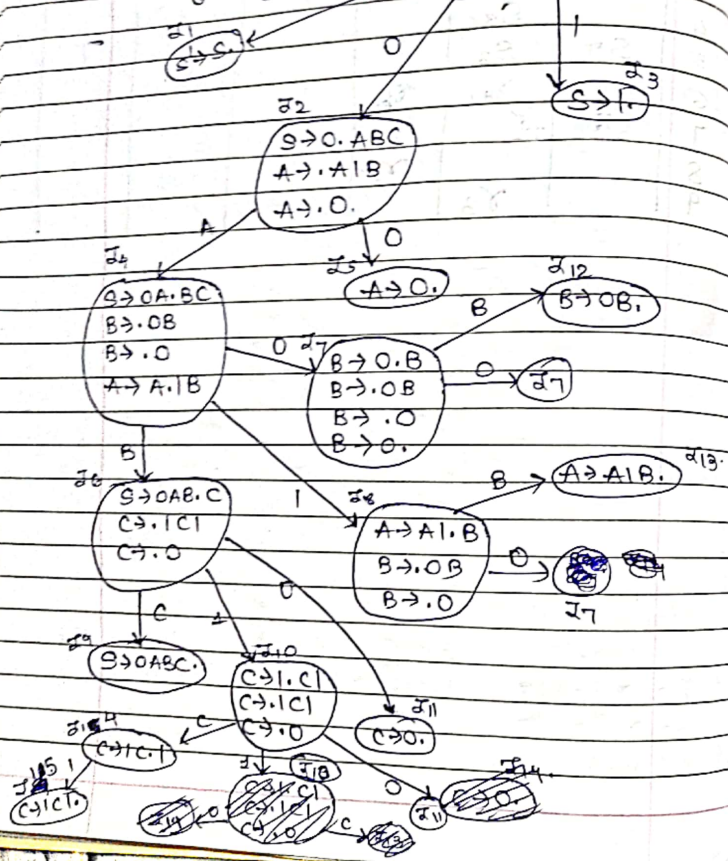
- ① $S \rightarrow aAB$
 - ② $S \rightarrow b$
 - ③ $A \rightarrow aA$
 - ④ $A \rightarrow b$
 - ⑤ $B \rightarrow b$
- $F(S) = \{a, b\}, \text{Foll} = \{\$ \}$
 $F(A) = \{a, b\}, \text{Foll} = \{b\}$
 $F(B) = \{b\}, \text{Foll} = \{\$ \}$

	a	b	\$	S	A	B
0	S ₂	S ₃		1		
1			Accept			
2	S ₅	S ₆			4	
3			r ₂			
4		S ₈				7
5	S ₉	S ₆			9	
6		r ₄				
7			r ₁			
8			r ₅			
9		r ₃				

Augmented Grammar

$S \rightarrow \epsilon ABC$
 $A \rightarrow A B$
 $B \rightarrow \epsilon B$
 $C \rightarrow \epsilon C$

$S' \rightarrow . S$
 $S \rightarrow . \epsilon A B C$
 $S \rightarrow . \epsilon$



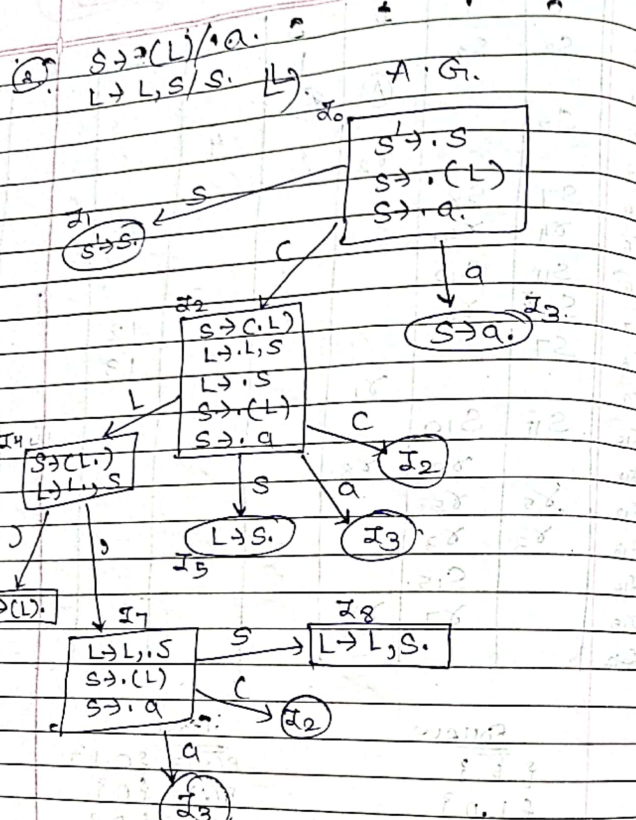
	0	1	\$	3	A	B	C
I ₀	S ₂	S ₃		1			
I ₁			(A)	(B)			
I ₂	S ₅				4		
I ₃			r ₂				
I ₄	S ₇	S ₈				6	
I ₅	r ₄	r ₄					
I ₆	S ₁₁	S ₁₀					9
I ₇	S ₇	r ₆				12	
I ₈	S ₇					13	
I ₉			r ₁				
I ₁₀	S ₁₁	S ₁₀					14
I ₁₁		r ₈	r ₈				
I ₁₂	r ₅	r ₅					
I ₁₃	r ₃	r ₃					
I ₁₄		S ₁₅					
I ₁₅		r ₇	r ₇				

Follow

- { \$ }
- { 1, 0 }
- { 1, 0 }
- { 1, \$ }

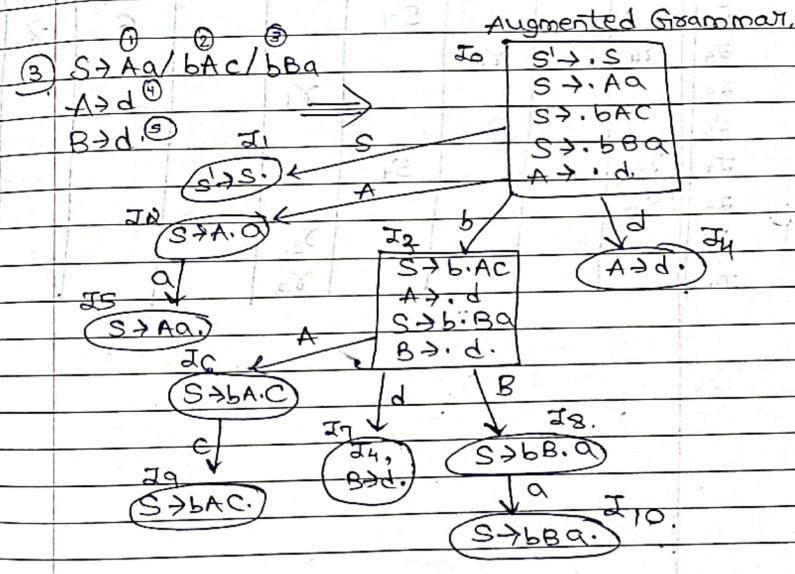
First

- $F(S) = \{ \epsilon, 1, 3 \}$
- $F(A) = \{ \epsilon \}$
- $F(B) = \{ \epsilon \}$
- $F(C) = \{ 1, 0 \}$



Follow: $\{ (,), a \}$
 First: $F(S) = \{ (, a \}$
 $F(L) = \{ (, a \}$

	C	)	,	a	\$	S	L
I0	S2			S3		1	
I1					(A)		
I2	S2			S3		5	4
I3		r2	r3		r2		
I4		S6	S7				
I5		r4	r4				
I6	1	r	r1		r1		
I7	S2			S3		8	
I8		r3	r3				



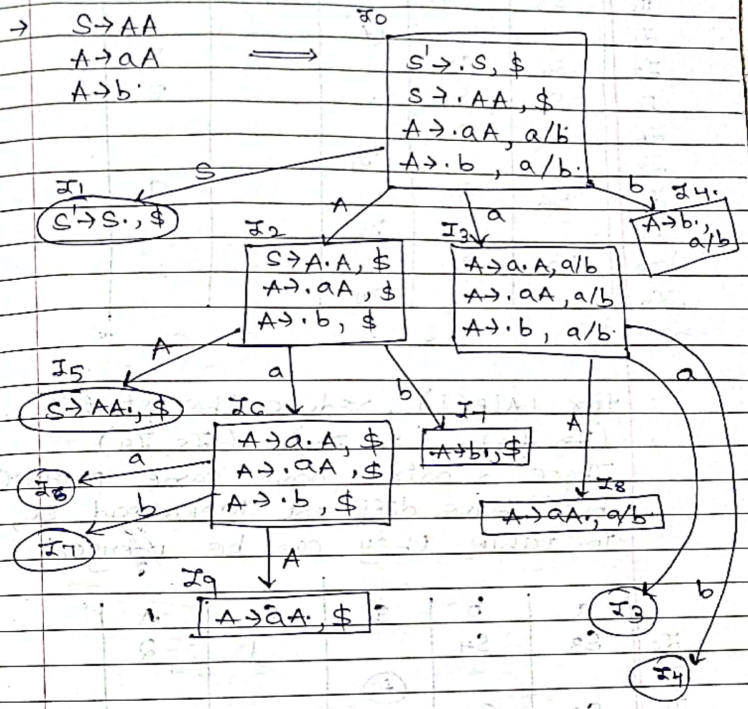
First

Follow
 $\{ \$ \}$ $\in S \rightarrow Aa/bac/bBa \Rightarrow F(S) = \{d, b\}$
 $\{a, c\}$ $\in A \rightarrow d \Rightarrow F(A) = \{d\}$
 $\{a\}$ $\in B \rightarrow d \Rightarrow F(B) = \{d\}$

	Action				State			
	a	b	c	d	\$	S	A	B
r0		S3		S4	(A)	1	2	
r1								
r2	S5							
r3				S7			6	8
r4	r4		r4					
r5					r1			
r6			S9					
r7	r5			S4				
r8	S6							
r9					r2			
r10					r3			

* CLR(1)

$\rightarrow S \rightarrow AA$
 $A \rightarrow aA$
 $A \rightarrow b$



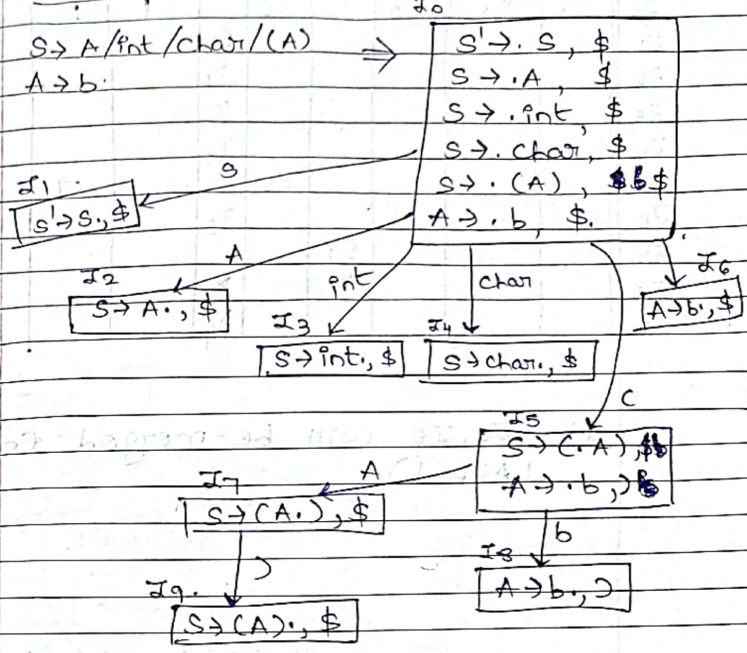
	a	b	\$	S	A
I ₀	S ₃	S ₄	(A)	1	2
I ₁					5
I ₂	S ₆	S ₇			8
I ₃	S ₃	S ₄			
I ₄	r ₃	r ₃	r ₁		9
I ₅					
I ₆	S ₆	S ₇			
I ₇			r ₃		
I ₈	r ₂	r ₂			
I ₉			r ₂		

For LALR(1), reduce the states:
 (I₃, I₆), (I₄, I₇), (I₈, I₉)
 These 3 pairs have same production
 but, have different lookahead so,
 in table they can be merged.

	a	b	\$	S	A
I ₀	S ₃	S ₄	(A)	1	2
I ₁					5
I ₂	S ₆	S ₇			8, 9
I _{3,6}	S _{3,6}	S _{4,7}	r ₃		
I _{4,7}	r ₃	r ₃			9
I ₅	S ₆	S ₇			
I _{8,9}	r ₂	r ₂	r ₂		

* LALR(1)

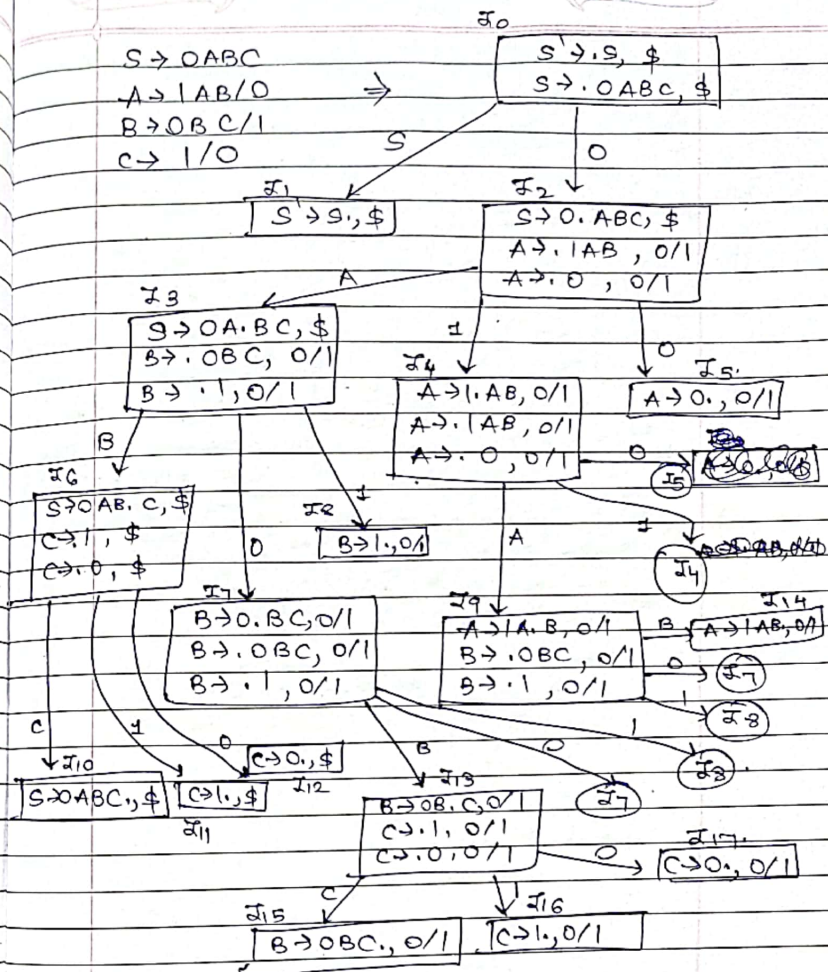
$S \rightarrow A / \text{int} / \text{char} / (A)$
 $A \rightarrow b$



	int	char	c	a	b	\$	9	1
I ₀	S ₃	S ₄	S ₅					
I ₁								
I ₂								
I ₃								
I ₄								
I ₅								
I ₆								
I ₇								
I ₈								
I ₉								

⊗ I₀, I₈ will be merged for LALP(1).

$S \rightarrow 0ABC$
 $A \rightarrow 1AB/0$
 $B \rightarrow 0BC/1$
 $C \rightarrow 1/0$



Accepts Ambiguous & Unambiguous grammar.

Date _____
Page 35.

⇒ Operator Precedence Parser :-

A Grammar 'G' does not contain

- ① Null Prod?
- ② 2 Adjacent variables on R.H.S. of prod?

$S \rightarrow SAS/a$ Not O.P.P.

$A \rightarrow bsb/b$ $S \rightarrow sbss/Sbs/a$
 $A \rightarrow bsb/b$

Q. $P \rightarrow SR/S$
 $R \rightarrow bSR/bS$
 $S \rightarrow WbS/W$
 $W \rightarrow L*W/L$
 $L \rightarrow id.$

Useless Prod.
So, remove it

$P \rightarrow SbSR/SbS/S$
 $R \rightarrow bSR/bS$

$P \rightarrow sbP/SbS/S$

$R \rightarrow bP/bS$

$S \rightarrow WbS/W$

$W \rightarrow L*W/L$

$L \rightarrow id.$

$P \rightarrow sbP/SbS/S$

$S \rightarrow WbS/W$

$W \rightarrow L*W/L$

$L \rightarrow id.$

$$E \rightarrow E + E / E * E / id$$

string:- $id * id + id \$$

Operator Relational Table

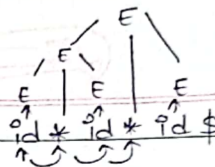
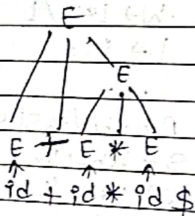
M	id	*	+	\$
id	-	>	>	>
*	<	=	>	>
+	<	<	=	>
\$	<	<	<	=

$id + id * id \$$

$\$ \mid id \mid + \mid id \mid * \mid id \mid$
X X X X X

Accept

IF precedence of P/P symbol is high $\Rightarrow$ Push or else $\Rightarrow$ Pop.



$\$ \mid id \mid * \mid id \mid + \mid id \mid$
X X X X X

Accept